

Exact values



1 Write down the exact value of the following:

a $\sin 60^\circ$

b $\cos 60^\circ$

c $\tan 30^\circ$

d $\sin 45^\circ$

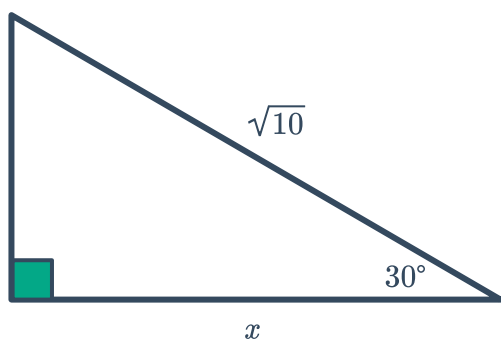
2 Given that $\sin \theta = \frac{1}{2}$ in a right triangle:

a State the value of θ .

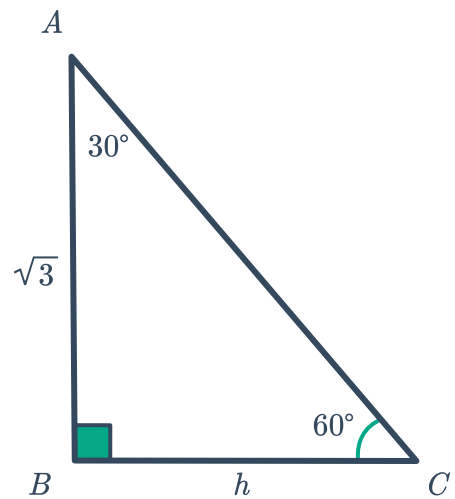
b Hence, find $\cos \theta$.

3 Find the exact value of the pronumeral(s) in the following triangles:

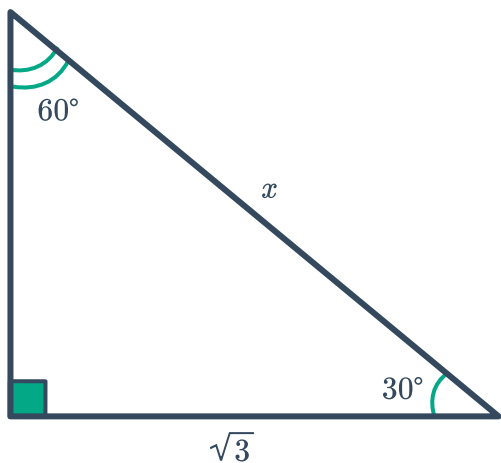
a



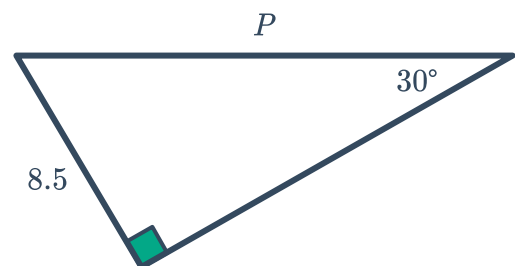
b



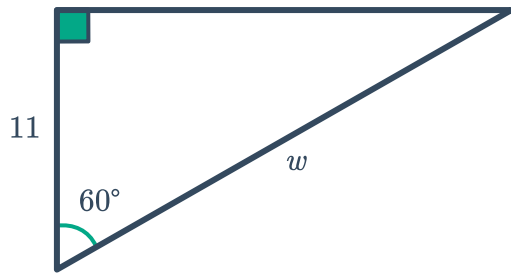
c



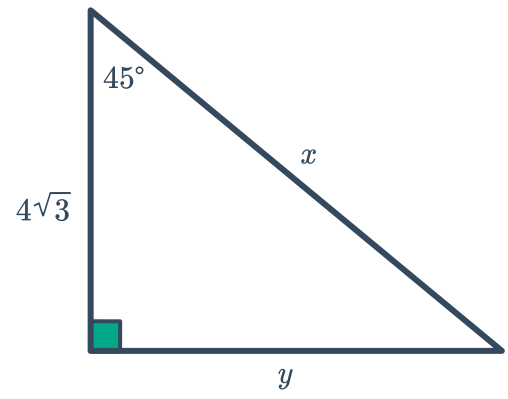
d



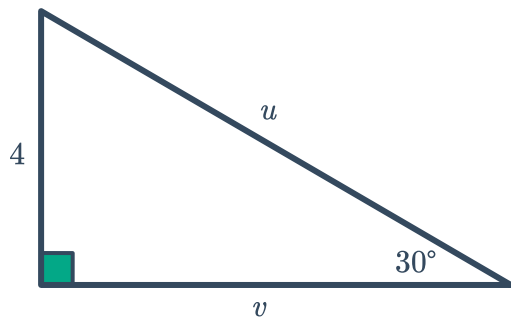
e



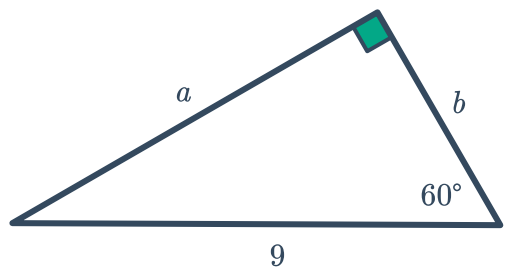
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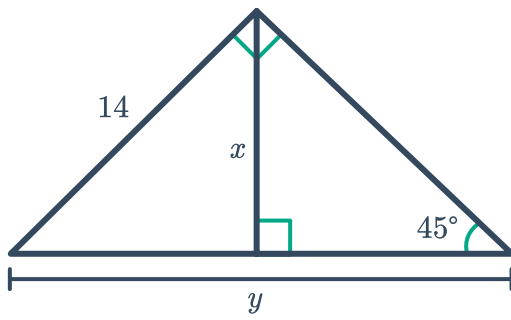
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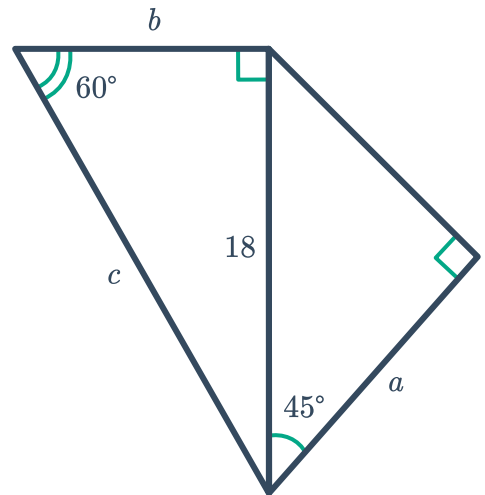
h



i

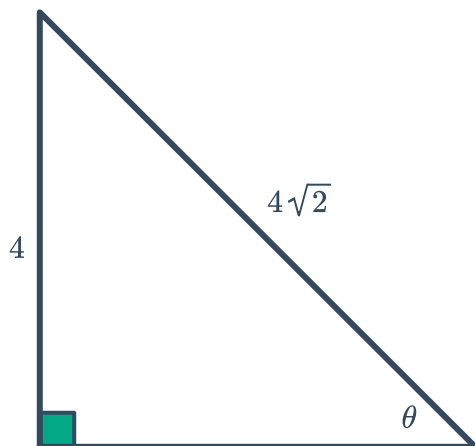


j

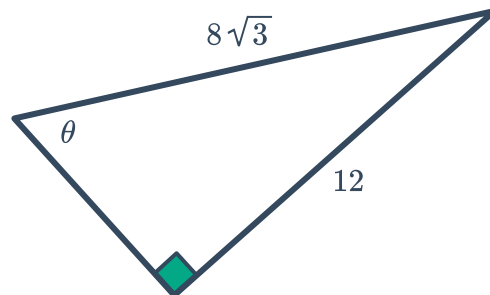


4 Use exact values to solve for θ in the following triangles:

a

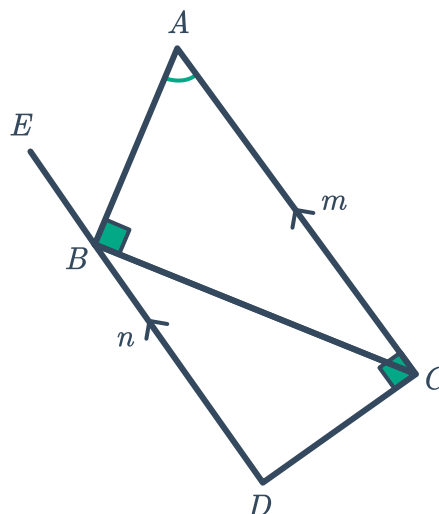


b



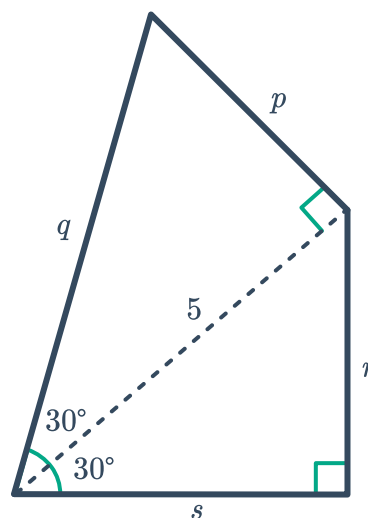
5 Consider the following diagram:

- a If $\angle EBA$ is 60° , name two other angles that are also 60° .
- b Given that CD has a length of $5\sqrt{7}$, calculate the exact values of m and n .



6 Consider the figure shown:

- a Find the exact length of the following:
 - i p ii q iii r iv s
- b Hence, find the exact perimeter of the quadrilateral.



7 Consider the given shape:



- a Find x .
- b Find the exact value of y .
- c Hence, find the exact perimeter of the shape.

